

HUM 4747: Legal Issues and Cyber Law

Non-Disclosure Agreements (NDA) & Trade Secrets

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Outline

- ① Introduction to Trade Secrets & NDAs
- ② Who Needs an NDA?
- ③ Key Components of an NDA
- ④ Confidentiality Obligations & Permitted Disclosure
- ⑤ Employee NDAs
- ⑥ Real-World NDA Example
- ⑦ What Should NOT Be in an NDA
- ⑧ NDAs in Bangladesh Context
- ⑨ Summary & Discussion

What is a Trade Secret?

A **trade secret** is any confidential business information that provides a company with a **competitive edge**.

Examples in Software Industry

Proprietary **algorithms** and source code

Software **architecture** and system designs

API keys, encryption methods, security protocols

Customer databases and user analytics

Business strategies, pricing models, product roadmaps

Think: Google's search algorithm, Coca-Cola's formula — but in software.

Why Do Trade Secrets Matter for Software Engineers?

As an Employee

You **will** handle confidential code, data, and designs

You may be asked to sign an NDA on your **first day**

Violations can lead to **lawsuits** and career damage

As a Founder / Freelancer

You need to **protect your own** innovations

Sharing ideas with investors or partners requires NDAs

Open-source vs. proprietary code decisions

What is a Non-Disclosure Agreement (NDA)?

Definition

A **Non-Disclosure Agreement (NDA)** is a legally binding contract between two or more parties that establishes a **confidential relationship**. The parties agree that certain information will **not be disclosed** to others.

Key Insight

An NDA is often the **first agreement** signed in any business-to-business or employer-employee relationship. It is imperative to negotiate a good NDA.

Types of NDAs

Unilateral (One-Way)

Only **one party** discloses confidential information

Common in **employer–employee** relationships

Example: A developer signs an NDA when joining a company

Mutual (Two-Way)

Both parties share confidential information

Common in **business partnerships** and joint ventures

Example: Two startups exploring a merger

*There can also be **multilateral NDAs** involving more than two parties.*

Who May Be Interested in an NDA?

Companies

Protect proprietary information

Safeguard trade secrets (formulas, designs, processes)

Clarify confidentiality policies to employees

Professionals

Sharing ideas with clients or partners

Disclosing sensitive info to service providers

Protecting early-stage prototypes

Individuals

Access to proprietary information of others

Disclosing ideas, plans, prototypes

Working with consultants or accountants

Software Industry Scenarios Requiring NDAs

1. **Hiring a developer / contractor** — Prevent code leaks
2. **Outsourcing to a vendor** — Protect system architecture details
3. **Pitching to investors** — Safeguard your business model
4. **Partnering with another company** — Sharing APIs or data
5. **Acquiring / merging with a company** — Due diligence phase
6. **Beta testing with users** — Unreleased product features
7. **Cybersecurity consulting** — Access to internal security posture

Eight Key Questions an NDA Should Answer

1. **Who** are the parties?
2. **What** is the scope of information?
3. **What type** of information will be exchanged?
4. **What media** will the data be in?
5. Are there any **exceptions**?
6. What are the **penalties** for breach?
7. **How long** will the NDA last?
8. Under what **law/jurisdiction**?

1. Identifying the Parties

Clearly identify all parties — names, designations, addresses

Are there more than two parties involved?

Will the NDA bind **subsidiaries**, **parent companies**, or **affiliates**?

Key Definitions

Disclosing Party The party sharing confidential information

Receiving Party The party receiving and protecting the information

Example

“This NDA is between *PricewaterhouseCoopers Bangladesh Pvt. Ltd.* (PwC) and *SECUPENT*, collectively the ‘parties’ ”

2–3. Scope and Type of Information

Scope — Don't Be Too Broad or Too Specific

The NDA should focus on a **particular program, proposal, or project**.

Types of Confidential Information (CI)

Technical: Source code, algorithms, system architecture, designs

Business: Pricing data, marketing plans, customer lists

Financial: Revenue projections, investor terms

Personnel: Employee data, organizational structure

“Any information marked as ‘Confidential’ or which is manifestly confidential in nature.”

4. Media of Exchange

How will confidential data be shared?

Written / Digital: Documents, emails, code repositories, databases

Oral / Verbal: Meetings, phone calls, presentations

Visual: Demos, prototypes, screen shares

Important for Oral Disclosures

If oral data is disclosed, how will each side **notify the other** that the information is considered proprietary and covered by the NDA?

*Typically: Disclosing party must provide a **written summary within 7 days** of oral disclosure, bearing a “Confidential” label.*

5. Exceptions — What is NOT Covered?

Information that is typically **excluded** from NDA protection:

1. Information already in the **public domain**
2. Information received from a **third party** not bound by confidentiality
3. Information **independently developed** by the receiving party
4. Information not properly **marked or qualified** under the NDA
5. Information the disclosing party shares **without restrictions**

Software Example

If a developer independently builds a feature similar to the one covered under NDA, they may not be in breach — provided they can **prove independent development**.

6. Penalties for Breach

What Happens if Someone Breaks the NDA?

Monetary damages — Compensation for financial loss

Injunctive relief — Court order to stop further disclosure

Indemnification — Breaching party covers all resulting costs

Termination of employment or business relationship

Criminal prosecution in severe cases (e.g., under trade secret laws)

Key Clause

"The misuse or unauthorized disclosure of CI would destroy or diminish the value of such information. Parties shall be entitled to remedies of injunction, specific performance and other equitable relief."

7–8. Duration and Governing Law

Duration

How long will the NDA last?

How long must parties protect data **after** the NDA expires?

Common durations: **1–5 years**

Some trade secrets: **indefinite**

Governing Law & Jurisdiction

Which country/state's law applies?

Which courts have jurisdiction?

Arbitration vs. **litigation**?

Cross-border considerations

Real Example

The PwC–SECUPENT NDA was governed by **Bangladesh law**, disputes resolved via **arbitration under the Bangladesh Arbitration Act 2001**, with Dhaka courts having jurisdiction. Duration: **1.5 years**.

How to Treat Confidential Information

Obligations of the Receiving Party

Keep all CI **confidential** and reasonably secure

Use CI **only** for the agreed purpose

Not disclose CI to any unauthorized third party

Apply **at least the same level** of protection as for own confidential data

Limit access to individuals on a **need-to-know basis**

For Software Engineers

This means: Don't copy proprietary code to personal repos, don't discuss architecture details on public forums, don't share API keys or credentials.

When Disclosure IS Permitted

Even under an NDA, disclosure may be allowed in certain circumstances:

1. With **prior written consent** of the disclosing party
2. On a **need-to-know basis** within the receiving party's organization
3. To **legal advisors** or **insurers** in case of disputes
4. When **required by law**, court order, or regulatory authority
5. To **subcontractors** — but they must sign equivalent NDAs
6. **Whistleblower protection**: Disclosing to attorneys or government officials regarding suspected violations of law is **not a breach**

Return and Destruction of Confidential Information

Upon Expiration or Termination

Return all CI and copies to the disclosing party

Destroy all notes, analyses, and documents derived from CI

Remove CI from all **digital storage** (servers, repos, local drives)

Certify in writing that all CI has been returned or destroyed

Ownership Reminder

CI disclosure is **NOT** a transfer of intellectual property rights.

“Any information, analyses, compilations, notes or other documents derived from or based on CI shall be and remain the property of the Disclosing Party.”

Purpose

Defines the circumstances under which an employee **can and cannot** disclose confidential information, including remedies and penalties.

Typical Structure of an Employee NDA:

1. **Scope of Agreement** — What's covered
2. **Definition of Confidential Information** — What counts as CI
3. **Exclusions** — What is NOT considered CI
4. **Period of Confidentiality** — During employment or indefinite
5. **Inventions Clause** — Who owns what you create
6. **Breach & Remedies** — Consequences of violation
7. **Governing Law** — Applicable jurisdiction

What Counts as Confidential Information?

In a software company context, CI typically includes:

Source code and algorithms

Software designs and blueprints

Computer programs, firmware

Research projects and prototypes

Formulae and techniques

Business development plans

Customer and investor data

Financial and pricing data

Marketing strategies

Employee information

Broad Definition

"Any material, knowledge, information and data concerning the Company or its businesses not generally known to the public...which, if misused or disclosed, could adversely affect the Company's business."

The Inventions Clause — Critical for Developers

Company-Owned Inventions

Anything you create **during employment**, using company resources, time, or information — belongs to the **company**.

Personal Inventions (Exceptions)

Inventions are yours if created:

- On your **own personal time**

- Not** on company property

- Without** using company materials, equipment, or information

Real-World Relevance

If you build a side project using your employer's laptop, cloud services, or drawing from knowledge of their proprietary systems — they may claim it!

Period of Confidentiality

Two common options:

Option A: Fixed Period

Employee agrees not to disclose CI for a specific period (e.g., **2 years after termination**).

Option B: Indefinite

Employee agrees not to disclose CI during employment **or any time thereafter**.

Tip for Software Engineers

Always read this clause carefully. An indefinite NDA means you can **never** discuss certain proprietary details — even years after leaving the company.

Case Study: PwC – SECUPENT NDA (Bangladesh)

Context

Parties: PricewaterhouseCoopers Bangladesh Pvt. Ltd. & SECUPENT

Purpose: WebTrust certification assessment for CCA, Root CA and Sub CAs in Bangladesh, plus cyber security assessment

Effective Date: September 11, 2019

Key Terms

Duration: 1.5 years

Governing Law: Laws of Bangladesh

Dispute Resolution: Arbitration under Bangladesh Arbitration Act 2001 (venue: Dhaka)

Damages: Restricted to **direct damages only**

PwC–SECUPENT NDA: Notable Clauses

Confidential Information Definition

Any information that is (i) marked “**Confidential**” in tangible form, OR (ii) confirmed in writing as confidential if disclosed orally, OR (iii) is **manifestly confidential**.

Oral Disclosure Protocol

Disclosing party must deliver a **written description within 7 days** of any oral disclosure, including date, place, and names of recipients.

Information Hierarchy

CI shared by SECUPENT was categorized by access level:

Senior Management: Board of Directors to Vice Presidents

Middle Management: General Managers to Managers

Junior Management: Assistant Managers and below

What Should NOT Be in an NDA

Do Not Include

Teaming Agreement details (work share, exclusivity)

Promises to perform on other contracts

Non-compete clauses (these belong in separate agreements)

Overly broad definitions that cover **all information ever shared**

Unreasonable durations (e.g., **lifetime NDAs for generic info**)

Best Practices

Always have **legal counsel** review before signing

Always know what you are signing!

Keep the scope focused on the **specific project or purpose**

NDA's Under Bangladesh Law

Legal Framework

Bangladesh does not have a **dedicated trade secret law**

NDA's are enforceable under the **Contract Act 1872**

ICT Act 2006 provides some digital information protections

Breach remedies available through **civil courts**

Arbitration commonly used (e.g., **Bangladesh Arbitration Act 2001**)

Functions of an NDA in Bangladesh

1. Define exactly what information **cannot** be disclosed
2. Protect **sensitive information** from being disclosed
3. Prevent **forfeiture of patent rights** (which may occur automatically upon public disclosure)

Key Takeaways

1. An NDA is **often the first legal document** you'll encounter in your career
2. Understand the **8 key questions** every NDA should answer
3. Pay special attention to the **inventions clause** and **duration**
4. Know the **exceptions** — not everything is covered
5. CI disclosure $\neq$ IP transfer
6. Always have legal counsel review; **always know what you're signing**
7. As future SWE professionals, you will both **sign and draft** NDAs

NDA Checklist for Software Engineers

Check Item	Status
Parties clearly identified?	☐
Scope focused on specific project?	☐
CI definition clear and not overly broad?	☐
Exceptions listed (public domain, independent dev)?	☐
Duration reasonable?	☐
Inventions clause reviewed?	☐
Breach penalties proportionate?	☐
Governing law and jurisdiction specified?	☐
Return/destruction of CI addressed?	☐
Reviewed by legal counsel?	☐

Discussion Questions

1. You're a freelance developer working on a client project. The client asks you to sign an NDA that covers “**all information ever discussed.**” What's wrong with this?
2. You build a side project on weekends using **similar technologies** to your employer's product. Could your employer claim ownership?
3. A startup founder shares their app idea with you and asks you to sign an NDA **before seeing the idea.** Should you sign?
4. How would you handle a situation where you need to **report a security vulnerability** you found in your company's software — but you signed an NDA?